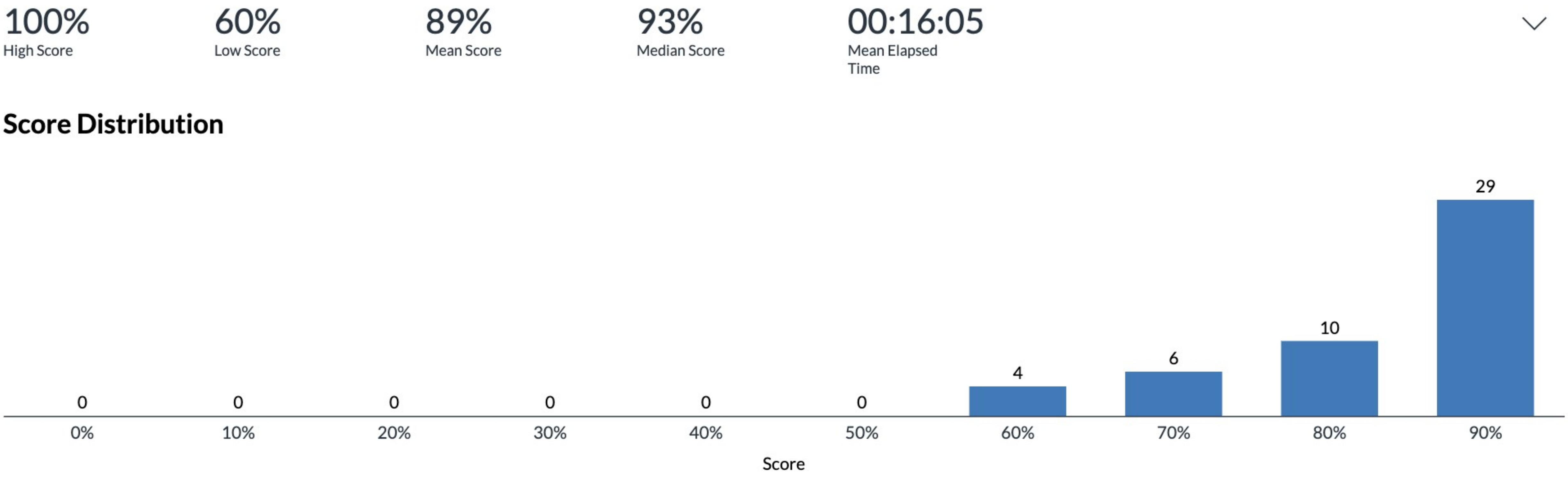
Week 10!?!

- Go over code beta test
- The “missing link” in our course

Coding quiz



100%
High Score
60%
Low Score
89%
Mean Score
93%
Median Score
00:16:05
Mean Elapsed Time
Score Distribution
0
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0
0
4
6
10
29
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
Score

When preparing training data for the DataLoader, we used something like `torch.stack([tokenize(n) for n in train_names])`. What does `torch.stack` do here?

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	Combines a list of individual tensors into a single batched tensor with a new first dimension	44	90%
✗	Concatenates all token tensors end-to-end into one long 1D tensor	3	6%
✗	Pads each tensor to the same length	2	4%
✗	Converts strings to tensors	0	0%

In the In-Context Learning (ICL) method, we included solved examples in the prompt. How were these examples structured in the messages list?

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	Examples were given as a pair of messages with questions from "role":"user" and answers from "role": "assistant"	38	78%
✕	Examples were all placed in a single markdown formatted table.	0	0%
✕	Examples were appended to the system prompt with instructions to answer in a manner similar to the provided examples.	11	22%

Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?

Sewon Min[1,2] **Xinxi Lyu**[1] **Ari Holtzman**[1] **Mikel Artetxe**[2]
Mike Lewis[2] **Hannaneh Hajishirzi**[1,3] **Luke Zettlemoyer**[1,2]
[1]University of Washington [2]Meta AI [3]Allen Institute for AI
{sewon,alrope,ahai,hannaneh,lsz}@cs.washington.edu
{artetxe,mikelewis}@meta.com

>2k cites

"other aspects of the demonstrations are the key drivers of end task performance:"

(1)the label space

(2) the distribution of the input text, and

(3) the overall **format of the sequence**

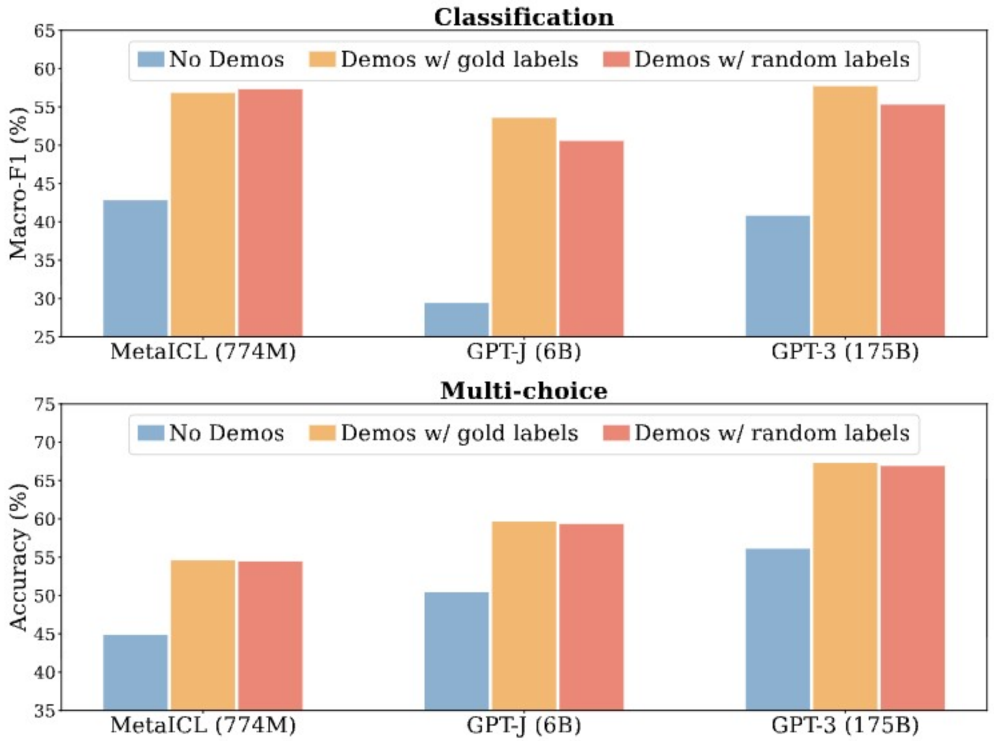


Figure 1: Results in classification (top) and multi-choice tasks (bottom), using three LMs with varying size. Reported on six datasets on which GPT-3 is evaluated; the channel method is used. See Section 4 for the full results. In-context learning performance drops only marginally when labels in the demonstrations are replaced by random labels.

Other observations on In Context Learning

- Formatting correctness means examples should be *exactly* like queries

- Increase number of examples?
  - Diminishing returns in accuracy
  - Increases cost – fills up context

- Larger models use context better

  https://arxiv.org/abs/2303.03846

  Larger language models do in-context learning differently

In HW1, the "base" method provided no instructions or examples. It got zero questions right (because it didn't know the alien words) but it was also one of the slowest methods to run, why?

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	Without formatting instructions it gave long responses	39	80%
✕	The model had to load additional weights for each problem	4	8%
✕	Base prompting disables GPU acceleration	0	0%
✕	The model retried each problem multiple times before giving up	6	12%

What is glorp(trok(trok(3, 4), flarn(1, 4)))? (max_token=256)

```
To solve the expression \(glop(trok(trok(3, 4), flarn(1, 4))) \), we need to break it down step-by-step. First, let's evaluate the innermost function: - The first argument of \( glop\) is \( trok(3, 4) \). - According to the problem statement, \( trok(3, 4) = 7 \). Next, we apply this result to the next function: - The second argument of \( glop \) is \( flarn(1, 4) \). - According to the problem statement, \(flarn(1, 4) = 5 \). Now, we combine these results using the outermost function: - The final argument of \( glop \) is \( \text{glop}(7, 5) \). According to the problem statement, \( glop 7, 5) = 20 \). Therefore, the value of \( glop (trok(trok(3, 4), fl
```

Cuts off mid-sentence

Observations:

- LLM = BS artist without context (makes up info, changes definitions)
- COT is default, format instructions needed to reduce tokens
- COT is verbose – if you stop early with “max_tokens” you won’t get to the answer

Temperature scaling divides logits by a scalar T before applying softmax. If a model is overconfident (ECE is high because probabilities are too extreme), which direction should you adjust T?

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	T > 1	42	86%
✗	T=1	1	2%
✗	T<1	6	12%
✗	T=0	0	0%

We trained with "label smoothing" to improve the calibration. How did we implement this?

Maybe not a good question?

- ***18% answer is kind of reasonable***
- ***We didn't implement anything from scratch so no reason to remember this***

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	By modifying a hyper-parameter in the cross entropy loss.	35	71%
×	We modified the labels by dividing by a temperature, T.	9	18%
×	We randomly flipped some of the labels.	1	2%
×	We added noise to some of the images.	4	8%

```
nn.CrossEntropyLoss(label_smoothing=0.1)
```

The Missing Link:
ML on Graphs

Slides from former PhD student:
Sami Abu-El-Haija
(Google Research -> vibe coding startup founder!?)

Why should you pay attention today?

M**ost** practical data is **relational**. **Examples don’t exist in isolation**.

- Users write (**NLP**) comments, upload (**Image**) photos, download apps, etc.
- While different models handle different modalities …
  - How can we combine information across these relationships?
  - ML models on Graphs, e.g., Graph Neural Networks.

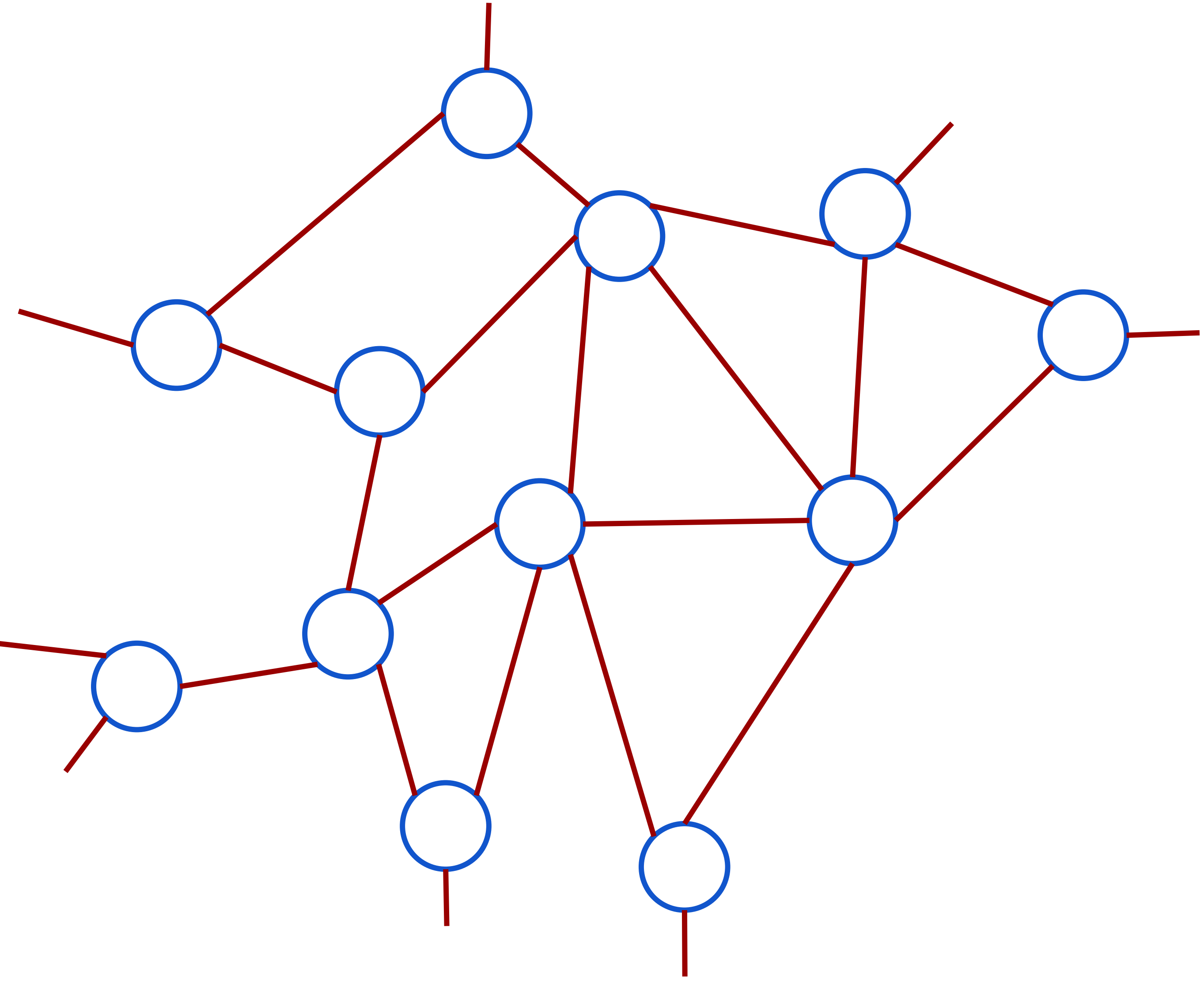
Agenda

- What is a Graph
- ML Tasks on Graphs
- Model Families for ML on Graphs
- Graph Neural Networks
  - Background
  - Practical Pitfalls (modeling & scale)
  - Addressing Pitfalls

Presentation prefers breadth over depth

What is a Graph

What is a graph?

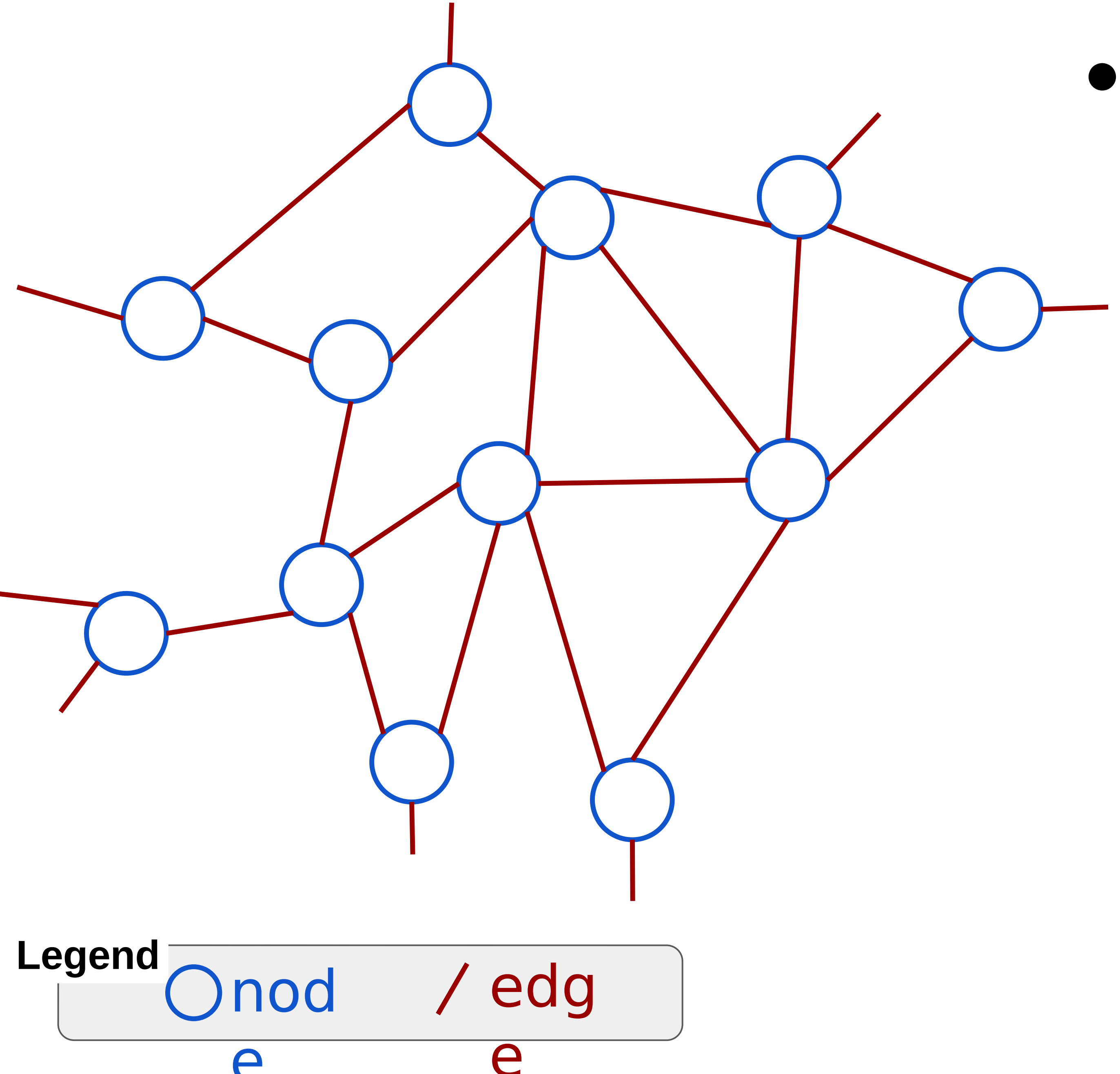


Legend

node

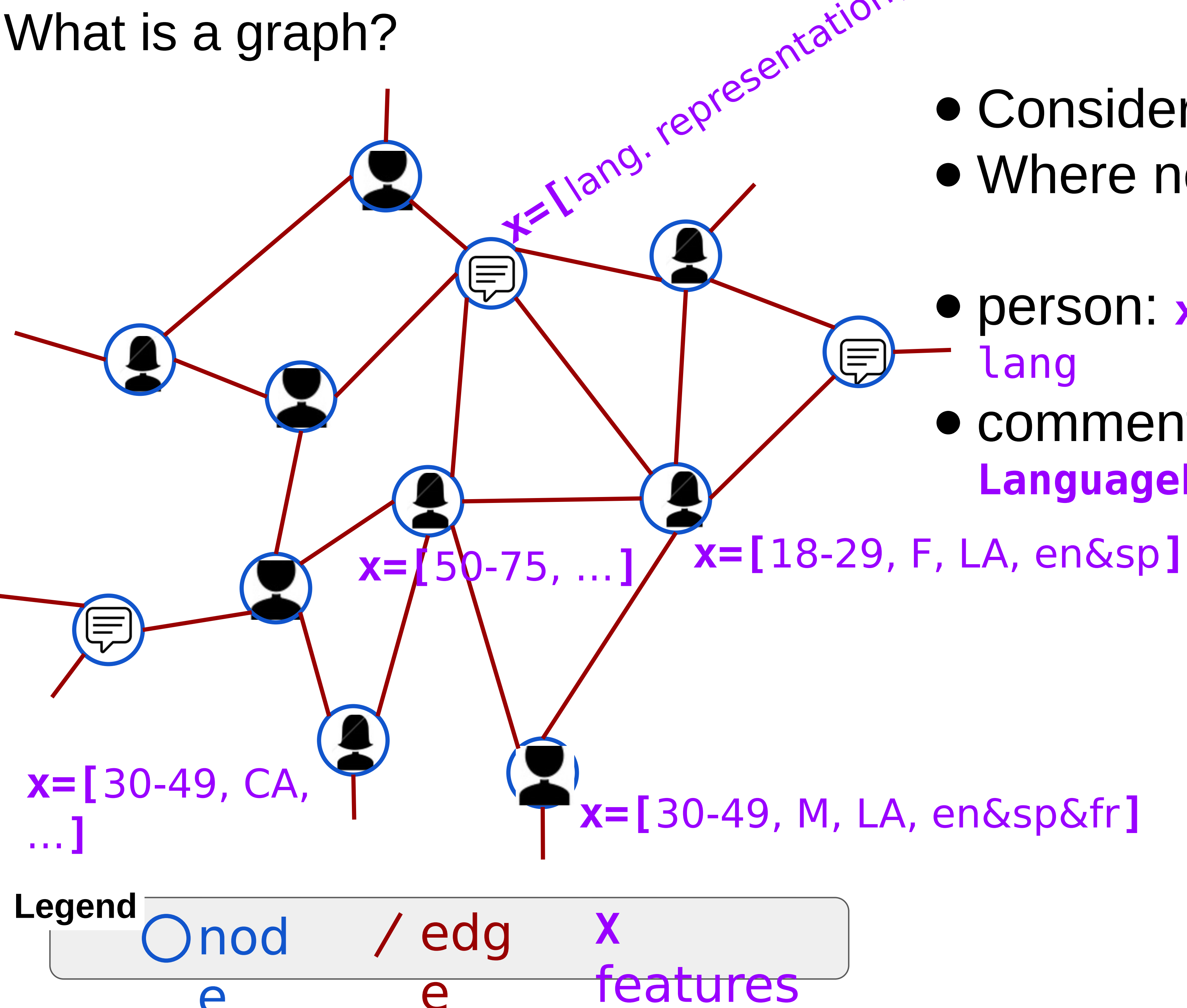
edge

What is a graph?



- Consider an online social network

What is a graph?



- Consider an online social network
- Where nodes can have ***features:***

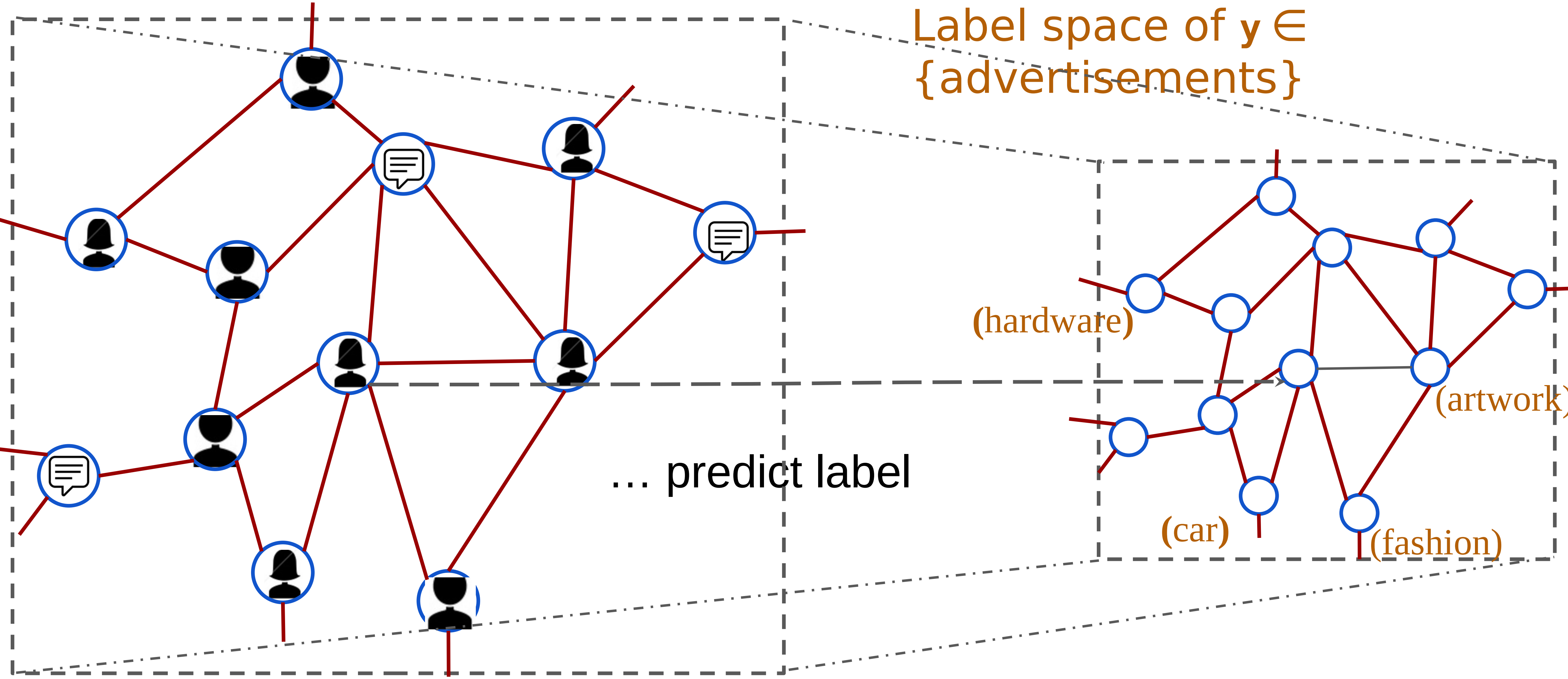
- person: `x = [age, gend, city, lang`
- comment: `x = LanguageEmbedding()`

ML Tasks on Graphs

Focus: Node Classification & Link Prediction

Machine Learning on Graphs can do **Node Classification**

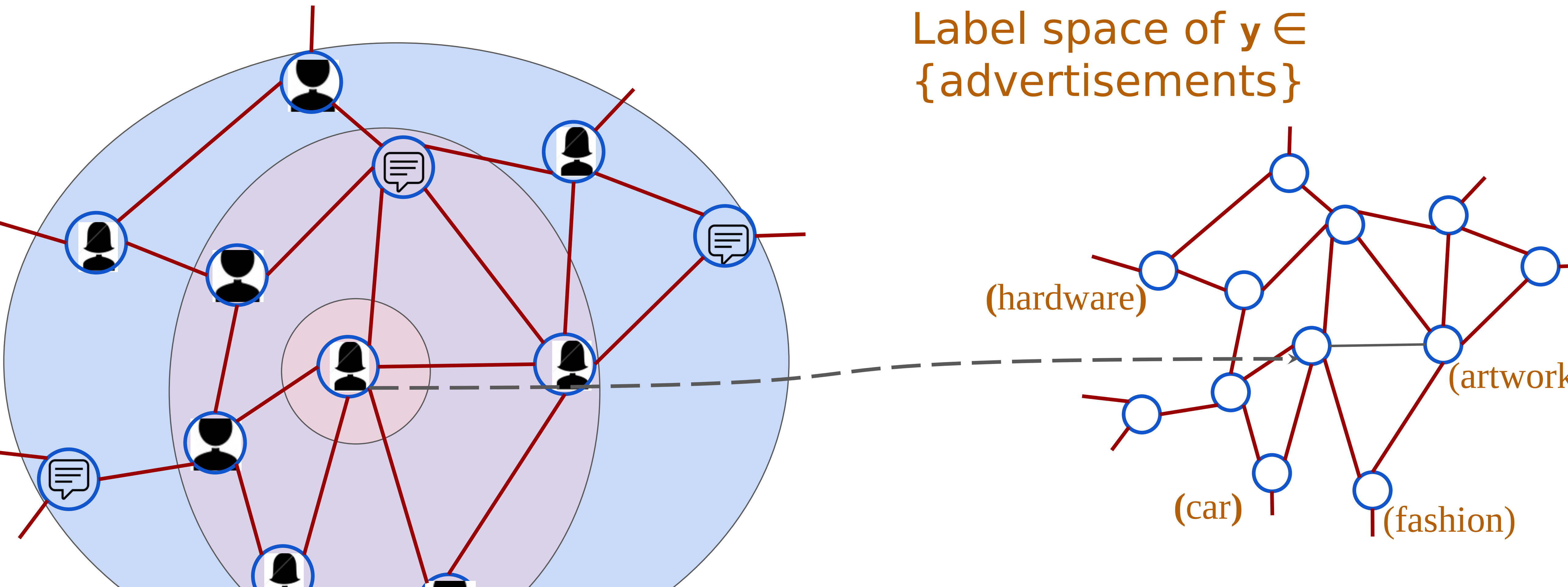
Machine Learning on Graphs can do **Node Classification**



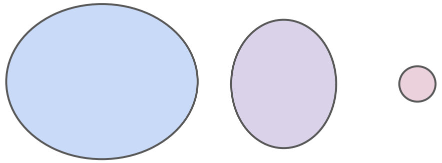
Legend

node / edge **x** features **y** label

Machine Learning on Graphs can do **Node Classification**



Using trainable patterns



Legend

nod
e

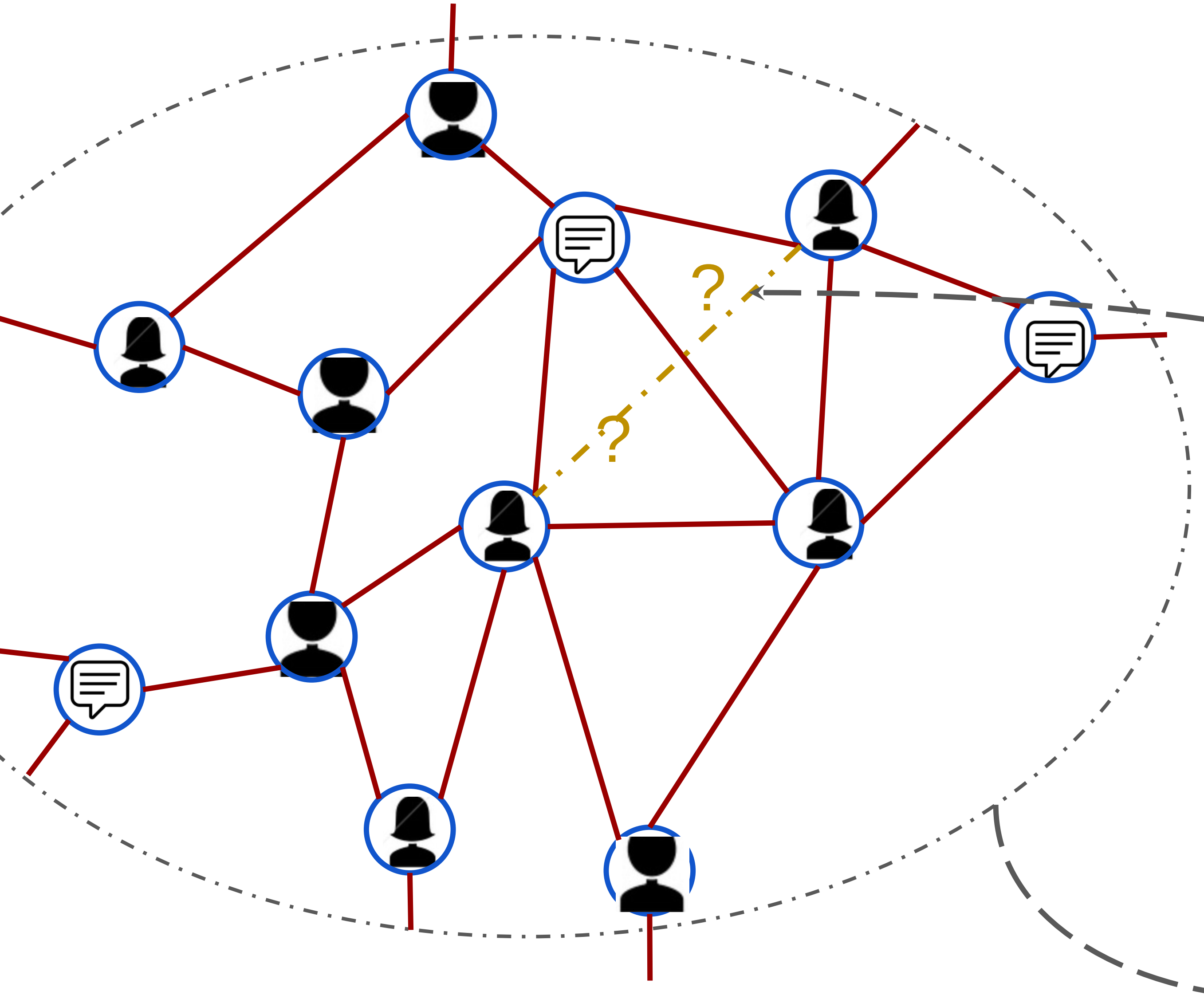
edg
e

x features

y label

Machine Learning on Graphs can do **Link Prediction**

Machine Learning on Graphs can do **Link Prediction**

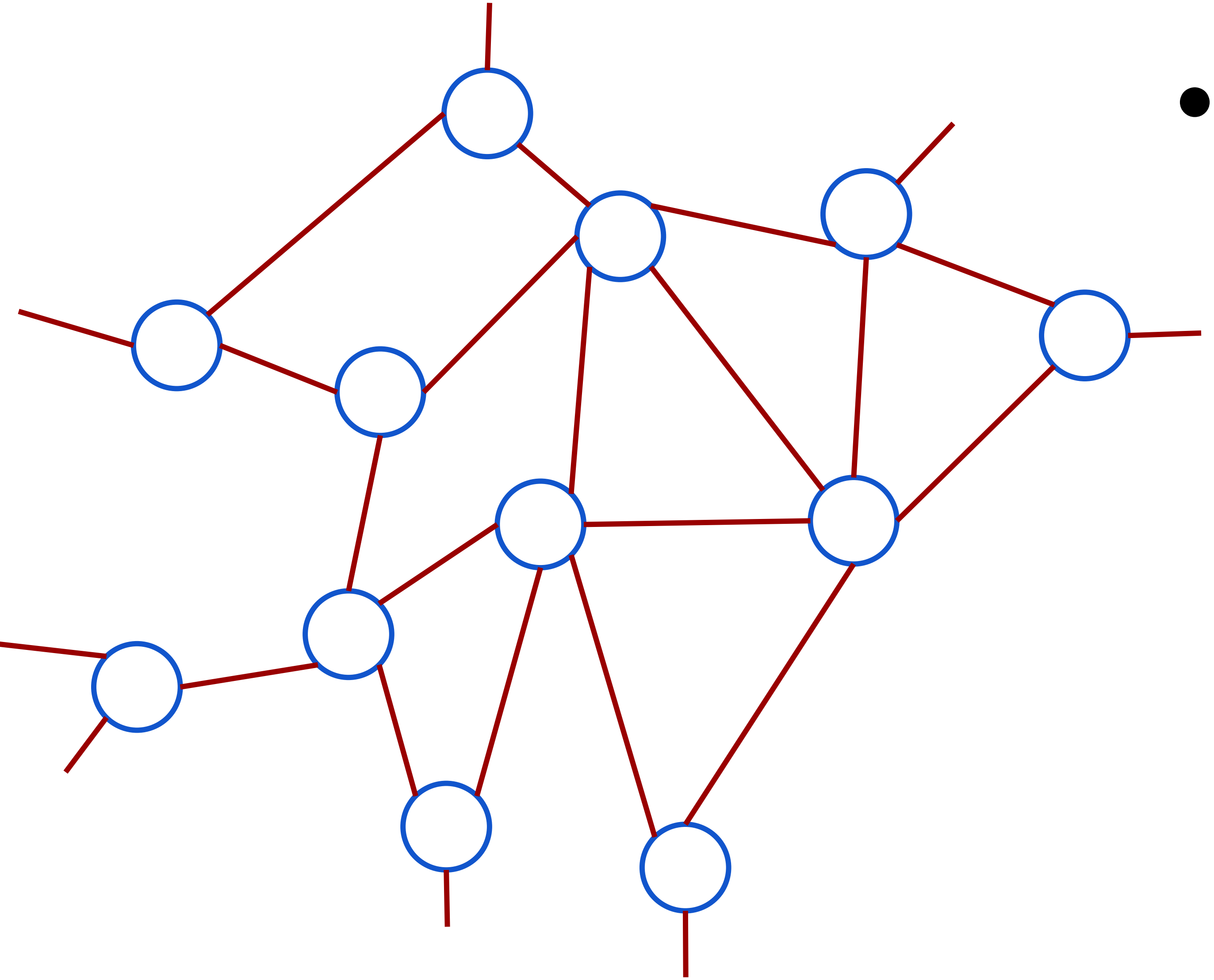


… e.g. for friend suggestion

Link prediction algorithm

Legend
nod e edg e X features y label

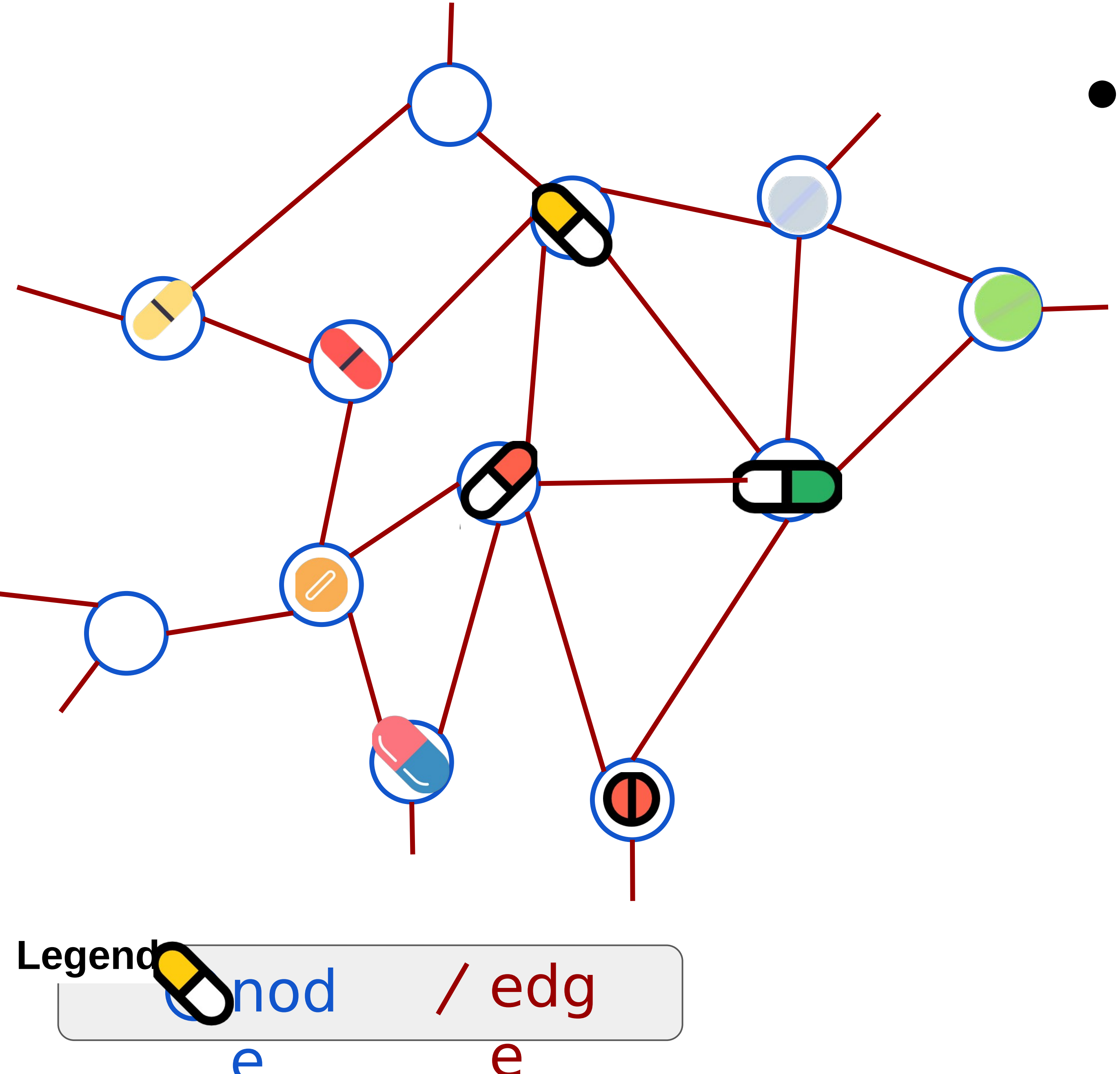
Link Prediction on drug-drug interaction networks



- Consider drug-drug interactions

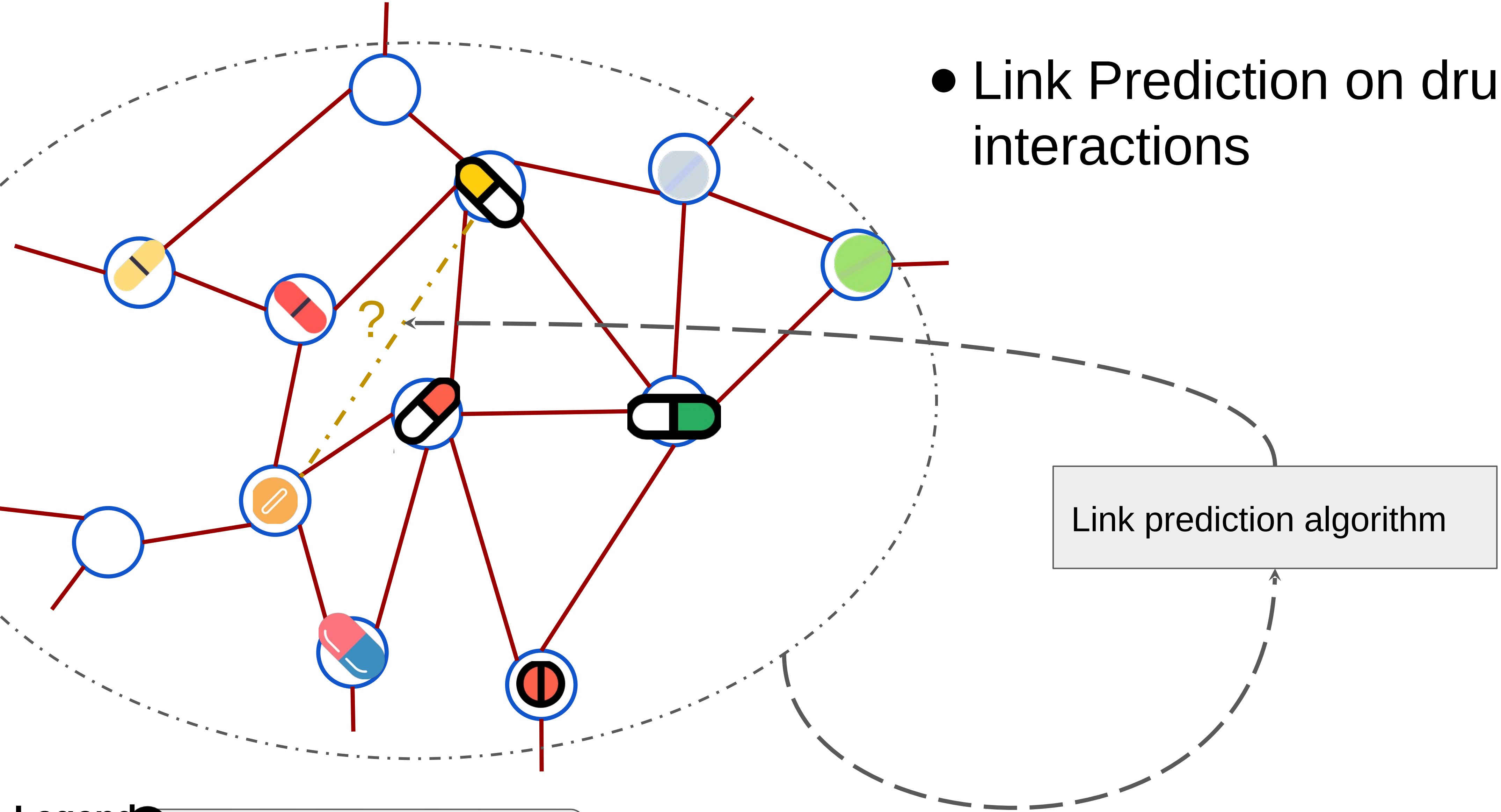
Legend

Link Prediction on drug-drug interaction networks



- Consider drug-drug interactions

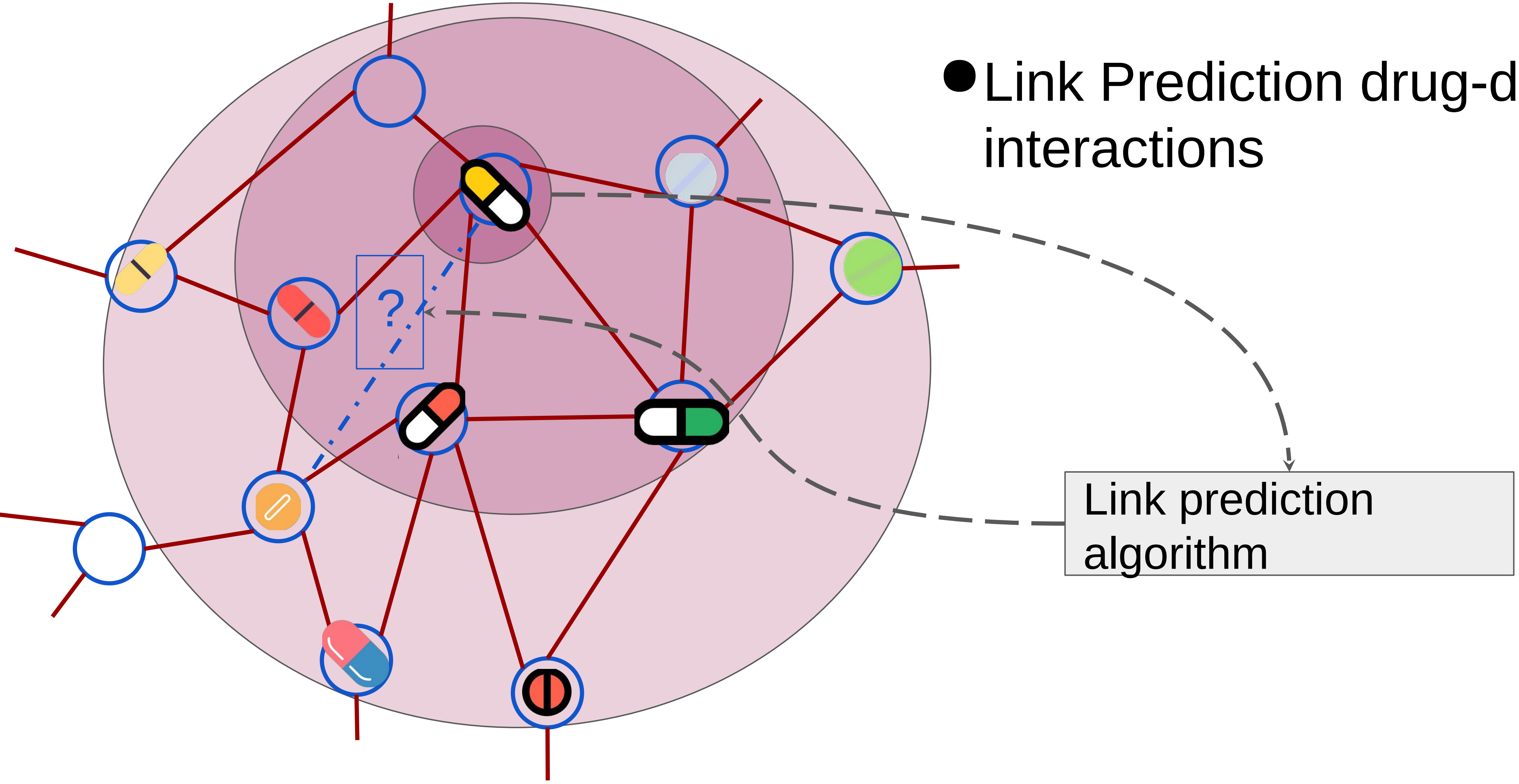
Link Prediction on drug-drug interaction networks



- Link Prediction on drug-drug interactions

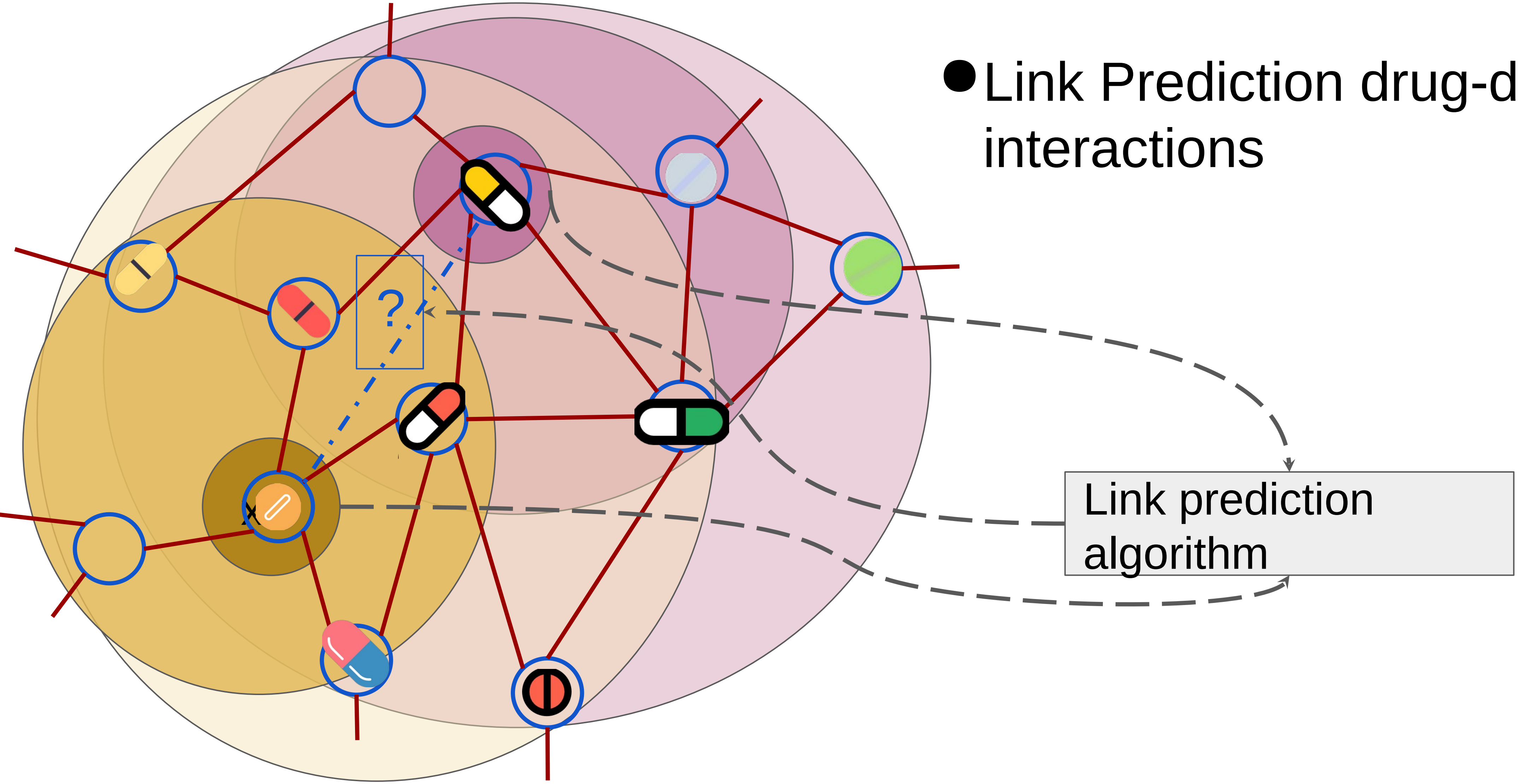
Link Prediction on drug-drug interaction networks

- Link Prediction drug-drug interactions

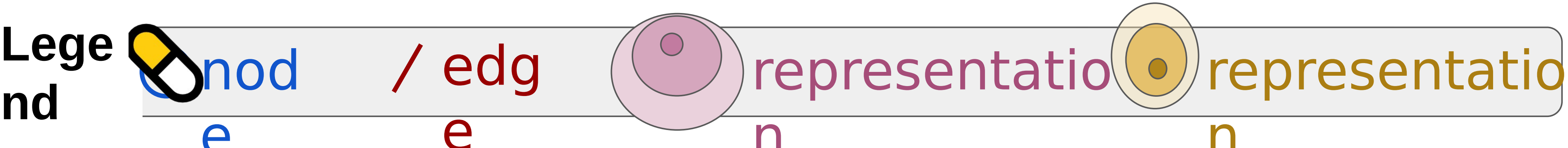


Lege nd nod e edg e representatio n

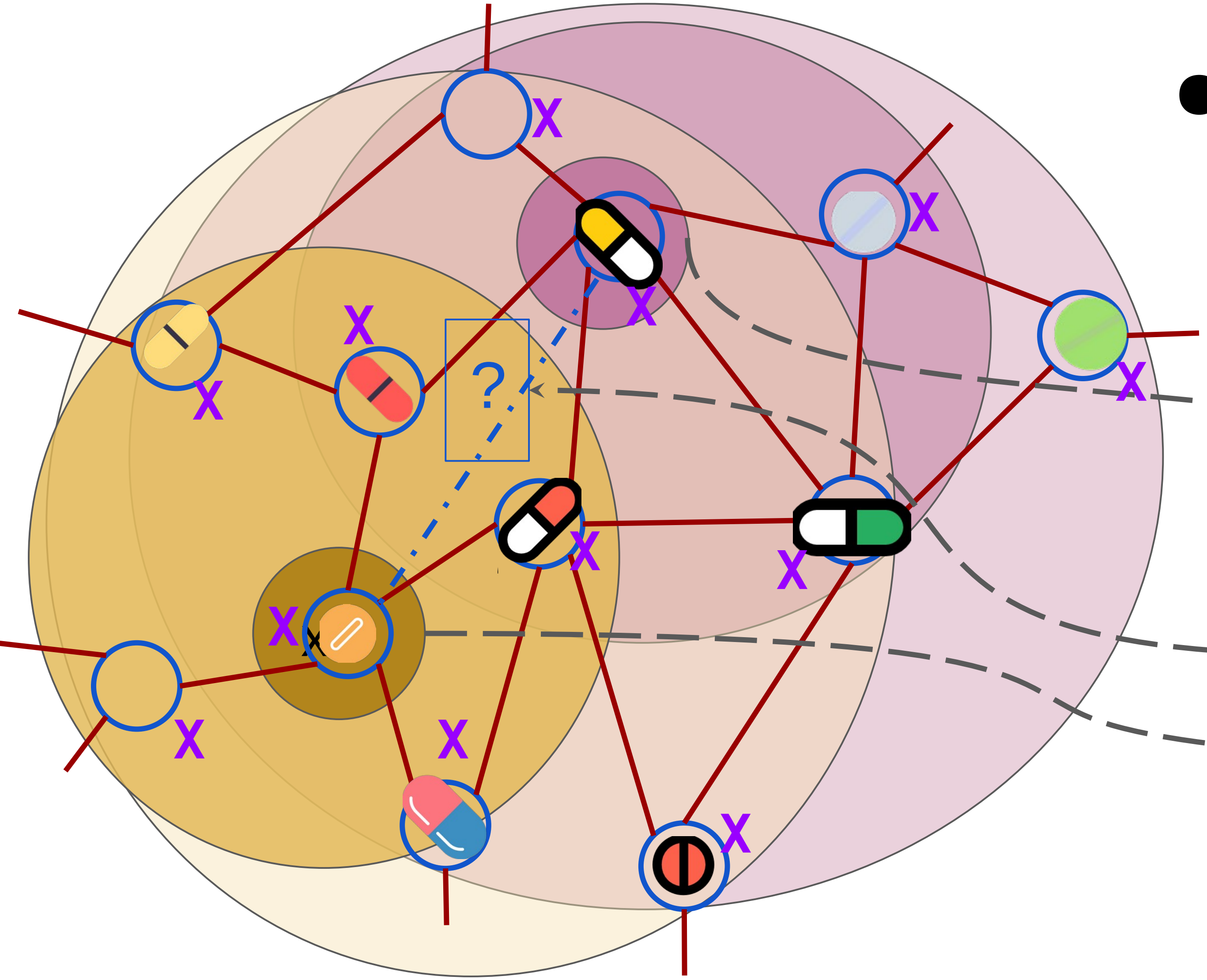
Link Prediction on drug-drug interaction networks



- Link Prediction drug-drug interactions



Link Prediction on drug-drug interaction networks



- Link Prediction drug-drug interactions

Link prediction algorithm

Legend: node, edge, representation, representation, X features

Why ML on Graphs (Motivation)

Graphs are general data structures. Algorithms are portable across domains.

- **Task** of *Link Prediction* and *Node Classification* apply to various domains.

Dataset	Node	Edge	Utility of Link Prediction	Utility of Node Classification

Why ML on Graphs (Motivation)

Graphs are general data structures. Algorithms are portable across domains.

- **Task** of *Link Prediction* and *Node Classification* apply to various domains.

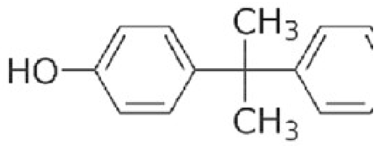
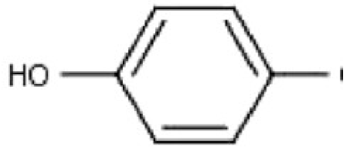
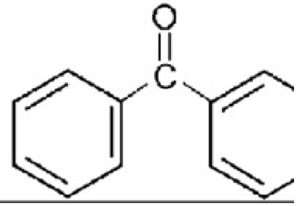
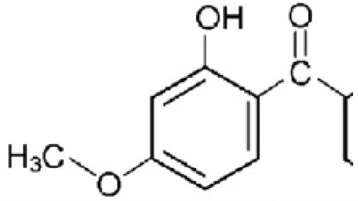
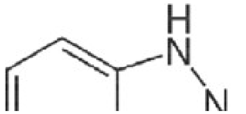
Dataset	**Node**	**Edge**	**Utility of Link Prediction**	**Utility of Node Classification**
Social Network	Person	Friendship	Friend Suggest	Ad preference
Netflix	User or Video	“Like”	Video Recommendation	What is this video about?
Drug-Drug interactions	Drug	interaction	Do 2 drugs interact?	Predict Drug bio{physical, chem} (e.g., toxicity, boiling temperature, antigen)
Citation Network	Article	Citation	Are articles related? (should one have cited the other)	Article Categorization

ML Tasks on Graphs

Beyond Node Classification & Link Prediction

ML on Graphs can do **Graph-level Prediction**

- Biochemistry

Compound	CAS number	Formula	Chemical Structure
Bisphenol A (BPA)	80-05-7	$C_{15}H_{16}O_2$	
Nonylphenol (NP)	104-40-5	$C_{15}H_{24}O$	
Benzophenone (BP)	119-61-9	$C_{13}H_{10}O$	
Benzophenone-3 (BP-3)	131-57-7	$C_{14}H_{12}O_3$	
Benzotriazole (BT)	95-14-7	$C_6H_5N_3$	

Task: output graph-level prediction, e.g., boiling temperature, binds to ProteinA?,..

- Computer Security
  - Computer programs are “Graphs”.
  - **Task**: Does a given program contains malware?

Summary of Feed-forward Tasks -- P(Y | Graph)

Input: Graph (nodes, edges, features)

Output:

- Graph-level (e.g., Graph Regression)
- Node-level (e.g., Node Classification)
- Node-pair level (e.g., Link Prediction)

inductive: test on new graphs

Can be **inductive**: test on a new graph

Can be **transductive**: test on training graph.

Can be on Homogeneous or Heterogeneous Graphs

Generative Tasks -- P(Graph | Y)

Input: Attribute

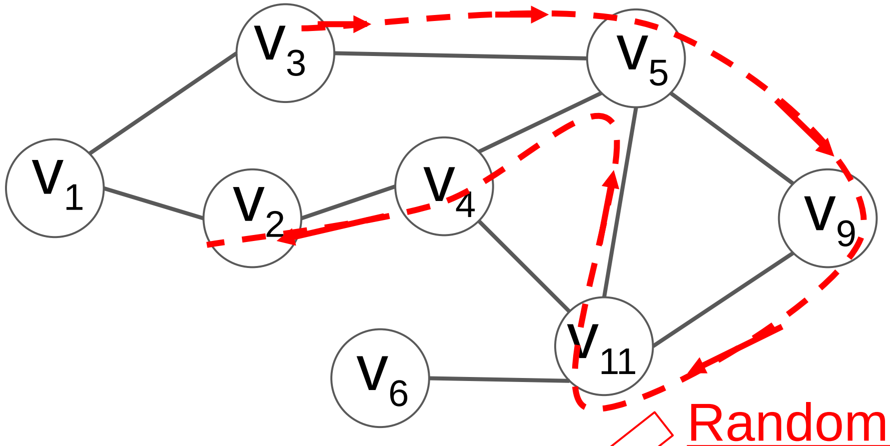
Output: Graph

E.g., drug synthesis

Will not cover today

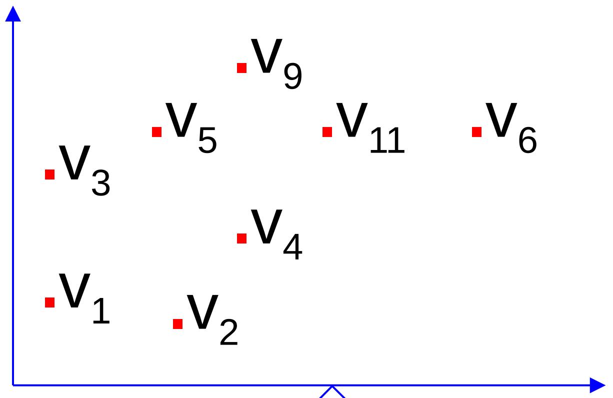
Model Family:
Skip-gram based
(most are transductive)

Skip-gram Based: embed each node



Embed in $\mathbf{R}^d$

Factorize **A** or **L**
Auto Encoder over **A**
Skipgram on E[walk] [2]



Random Walk

$V_3 \rightarrow V_5 \rightarrow V_9 \rightarrow V_{11} \rightarrow V_5 \rightarrow \ldots$

…

Random Walk Sequences

NLP embed alg e.g. word2vec

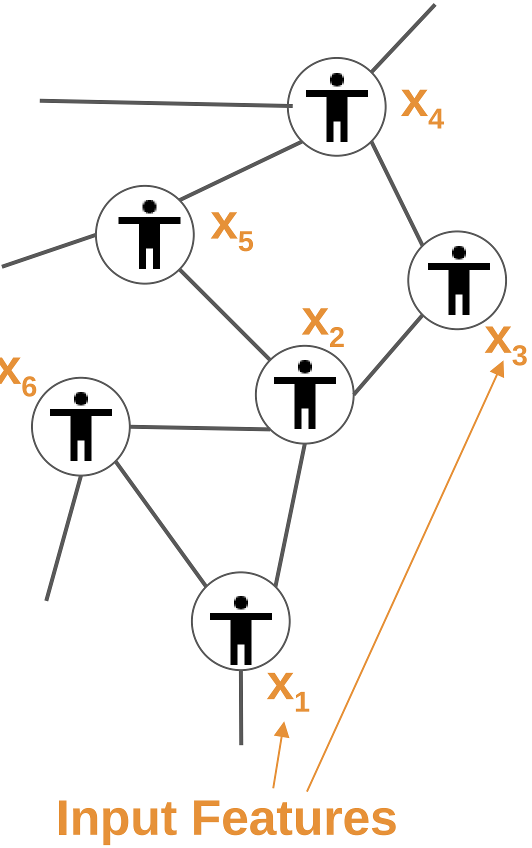
Perozzi et al, KDD 2016

X node features (n, d); **A** adjacency matrix (n, n)

X_i feature for node i $A_{i,j}$ = 1 iff i-j are connected

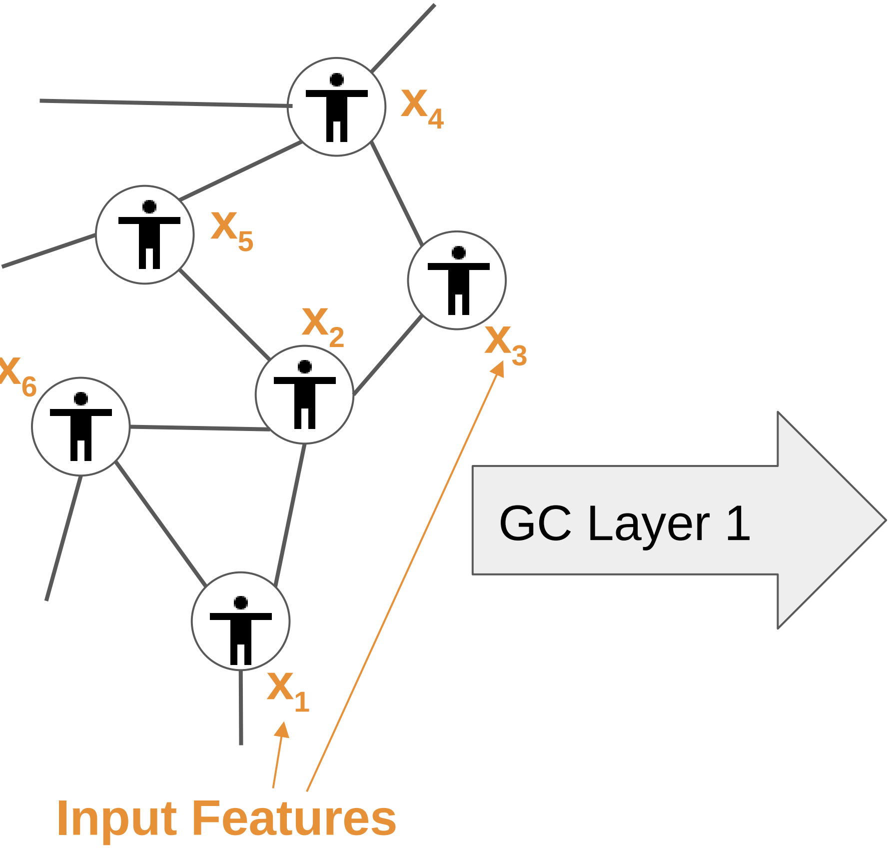
Model Family:
<u>Message-passing based</u>
(most are inductive)

Graph Convolutional Network (GCN)



Kipf & Welling, ICLR 2017

Graph Convolutional Network (GCN)



Kipf & Welling, ICLR 2017

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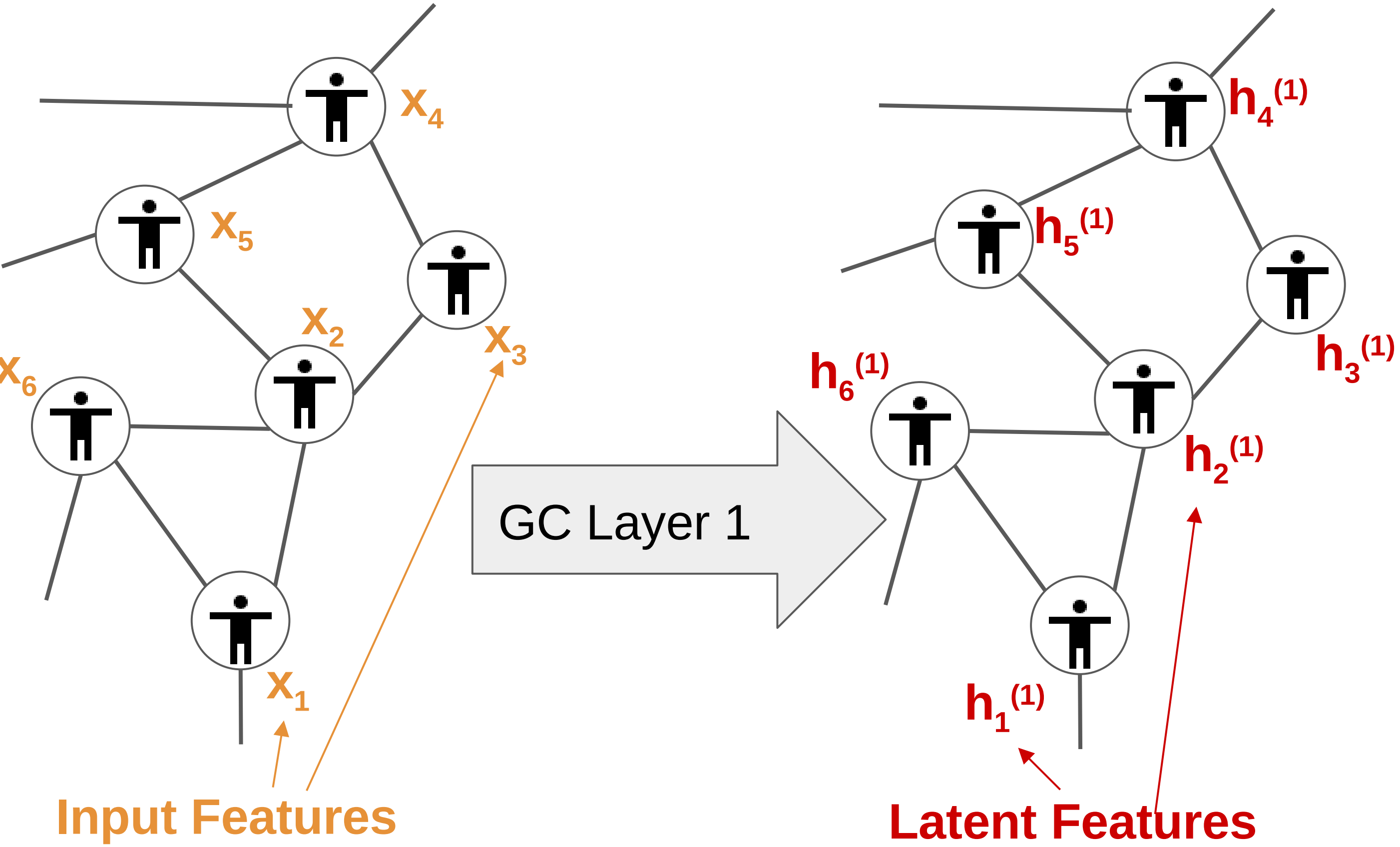

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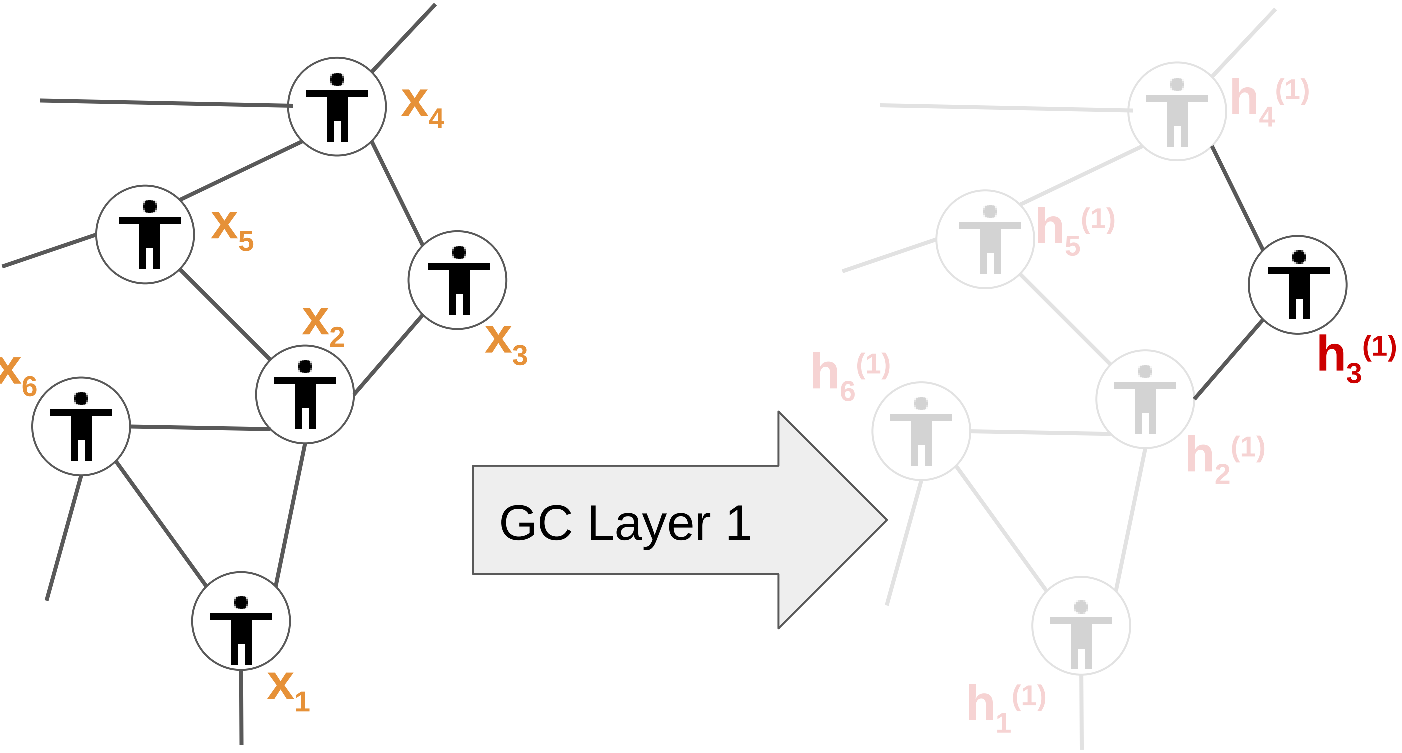

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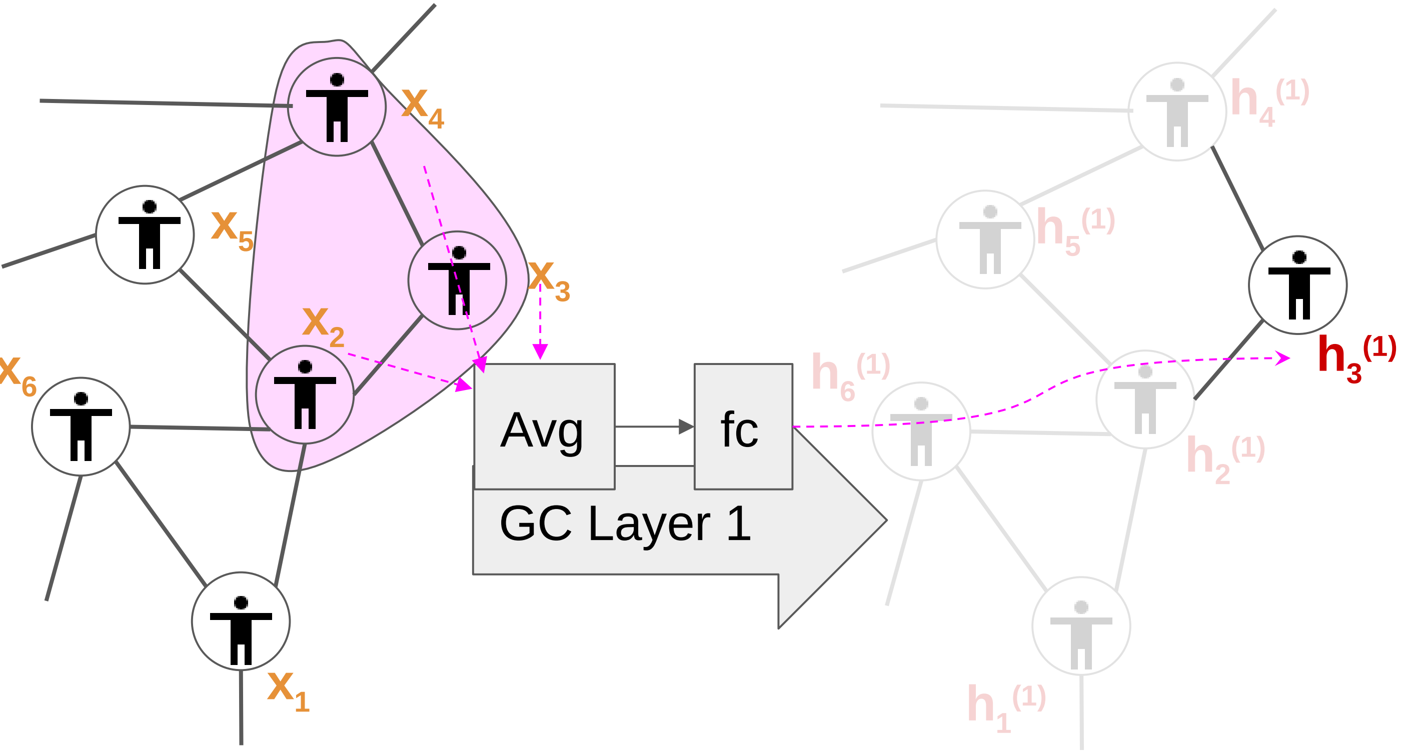

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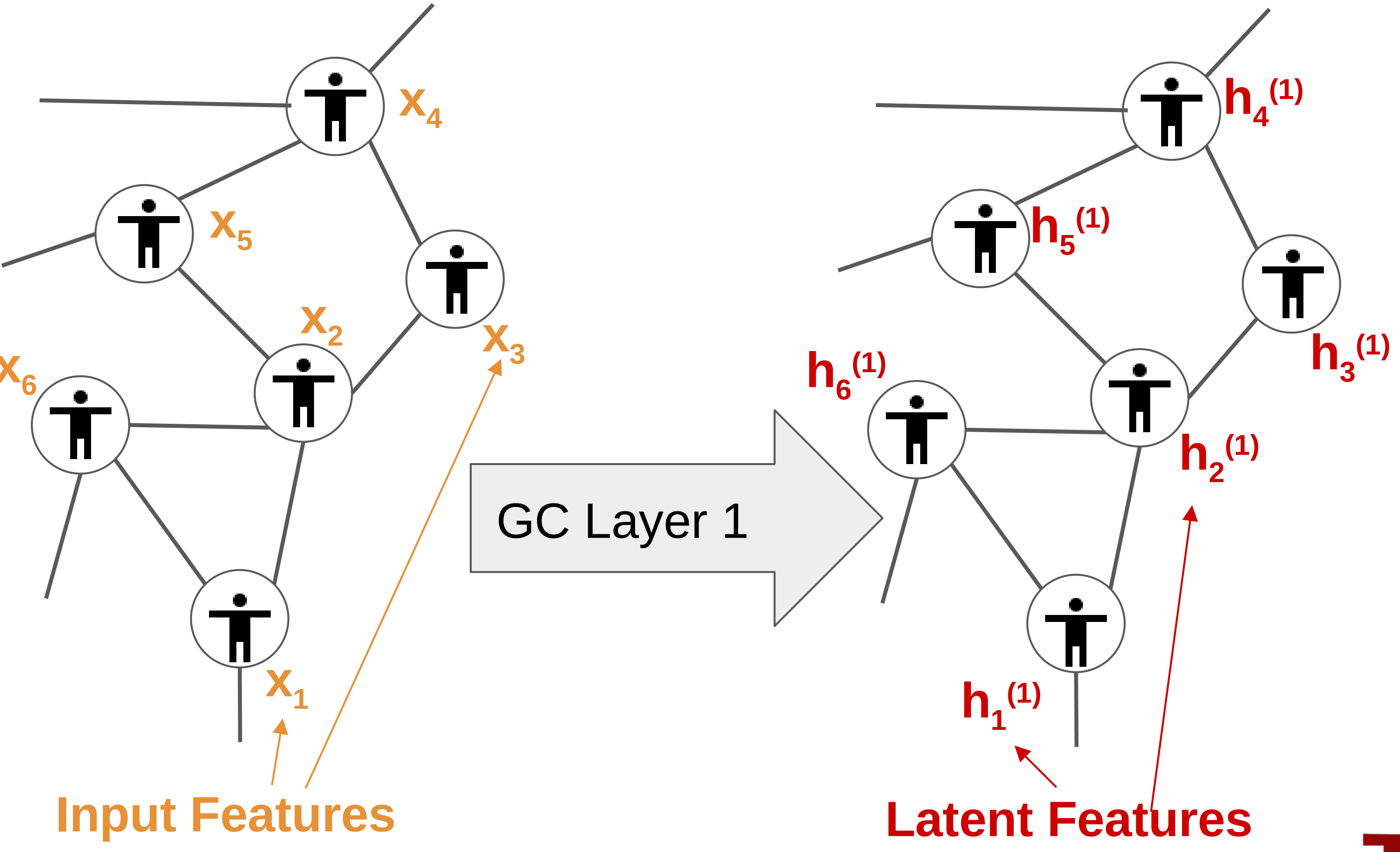


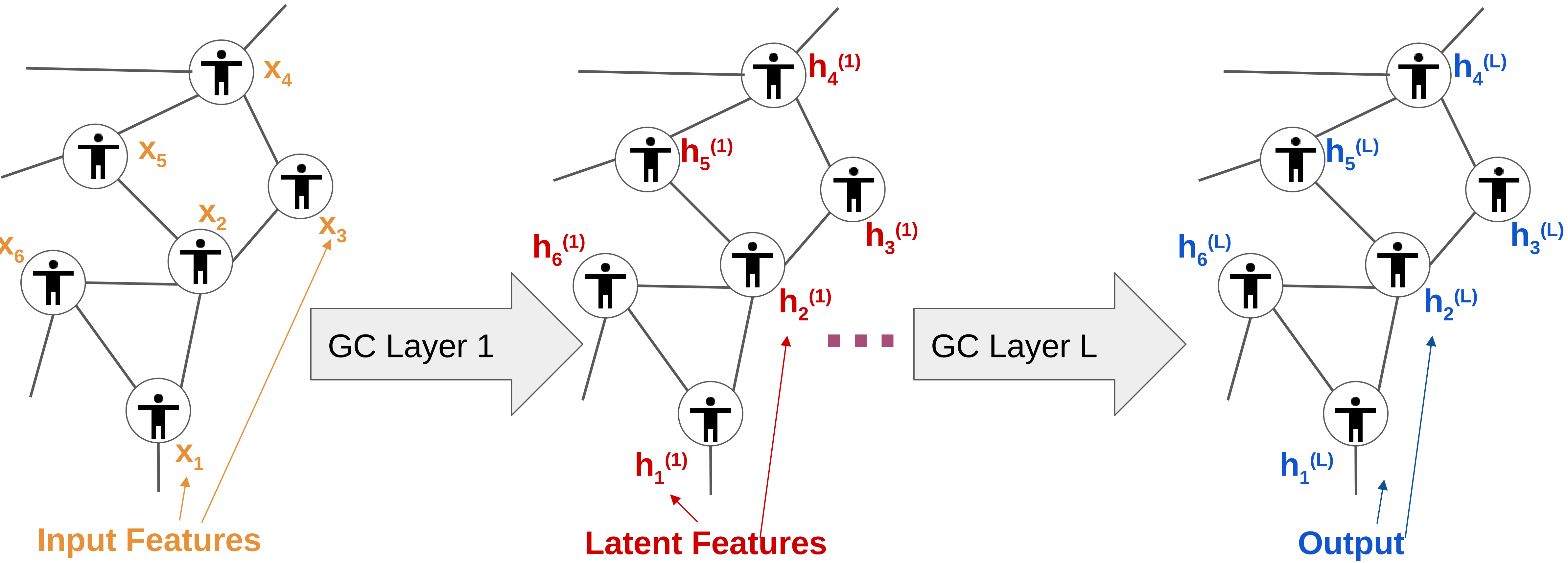
Note: Averages every node features with its 2-step neighborhood

$$\hat{A}^2 X$$

$$H^{(1)} = \mathrm{fc}(\overbrace{\hat{A}X}^{\text{Averages every node's features with its neighbors}})$$

Kipf & Welling, ICLR 2017

Graph Convolutional Network (GCN)



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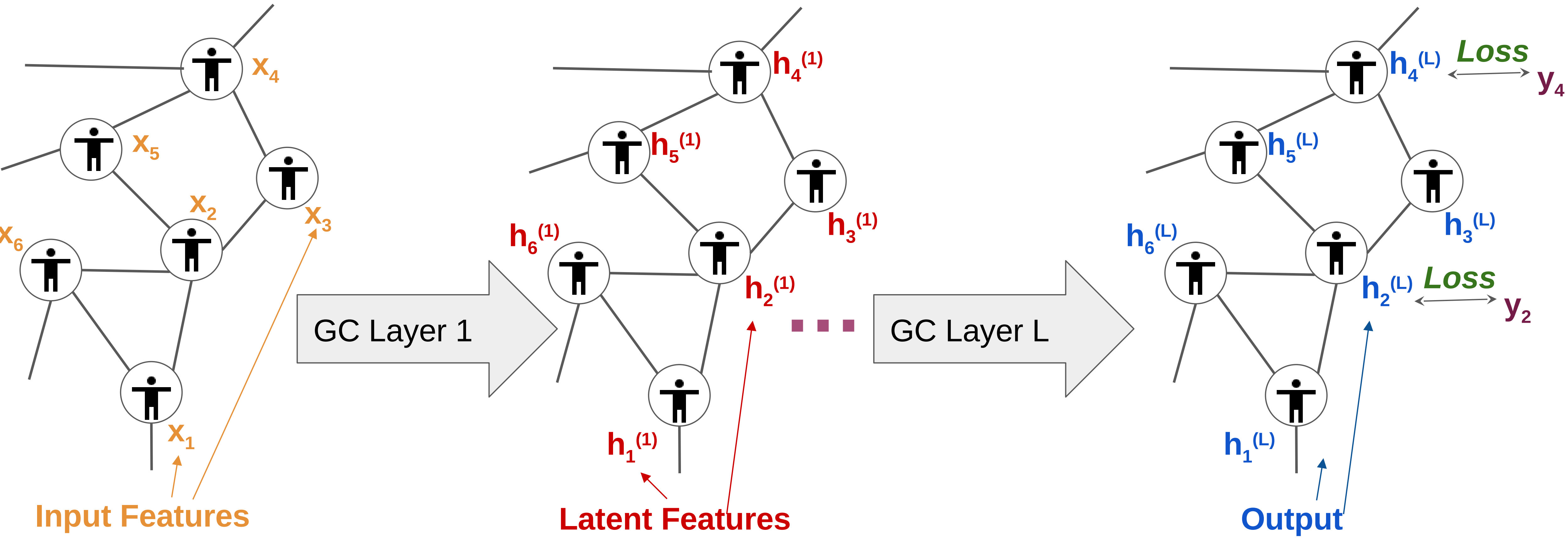

Kipf & Welling, ICLR 2017

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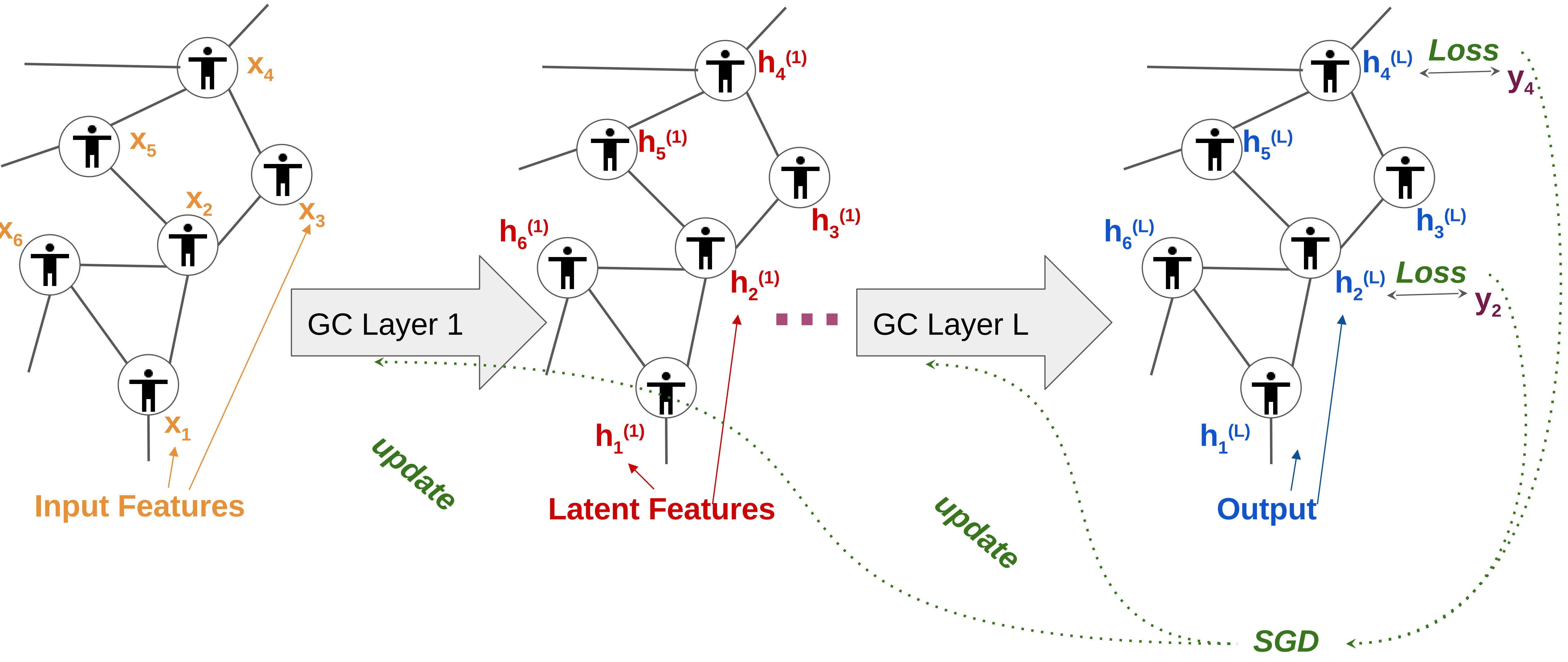


Kipf & Welling, ICLR 2017

Pitfalls

- Modeling Capacity
  - Adding more GNN layers “over-smoothes” the representation.
    - With many layers, each node will have the exact same representation **(eventually averages whole graph)**

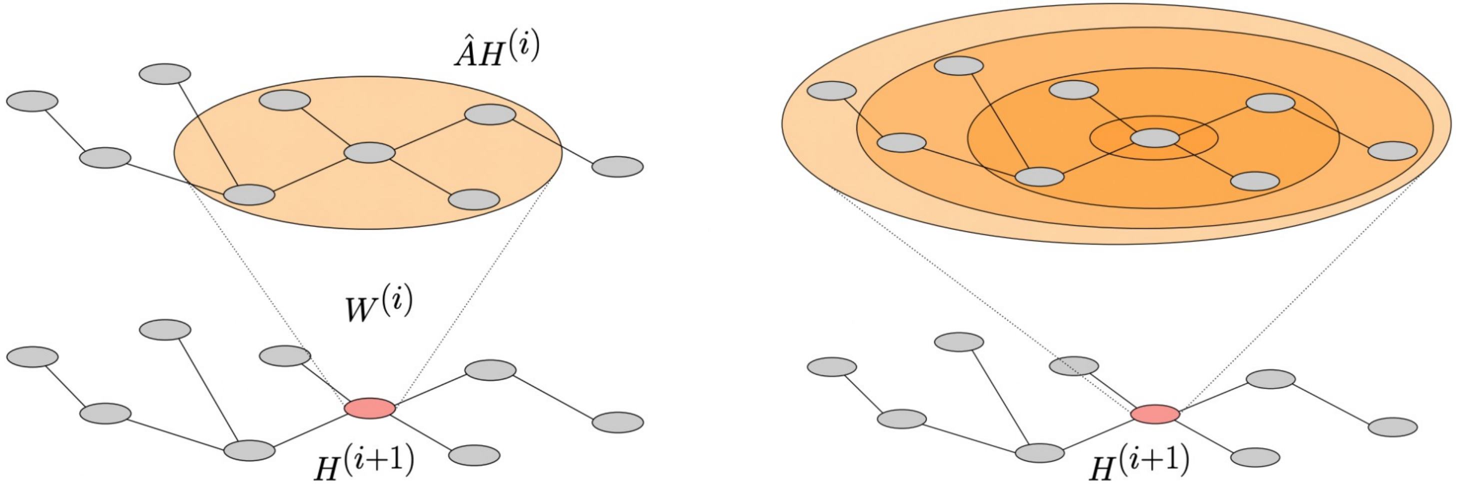
- Scalability
  - Early GNNs are formed on the graph as a whole (**No iid samples – no way to get batches?)**

Addressing Pitfall #1: Modelling Capacity

International Conference on Machine Learning, 2019

MixHop: Higher-Order Graph Convolutional Architectures via Sparsified Neighborhood Mixing

Sami Abu-El-Haija [1] Bryan Perozzi [2] Amol Kapoor [2] Nazanin Alipourfard [1] Kristina Lerman [1]
Hrayr Harutyunyan [1] Greg Ver Steeg [1] Aram Galstyan [1]



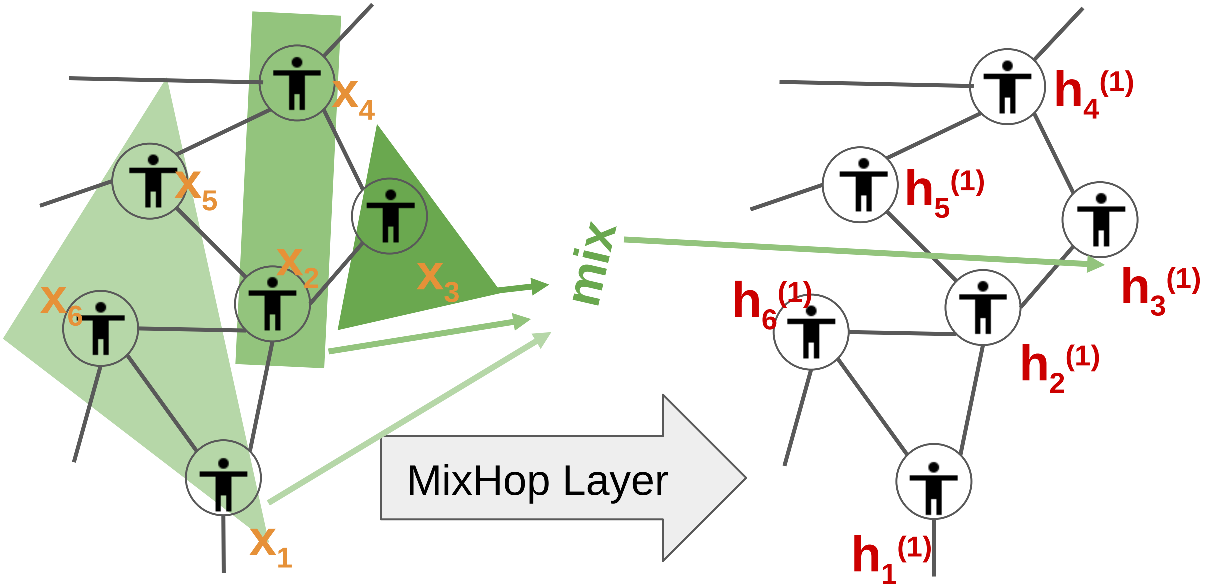
$H^{(i+1)} = \sigma(\hat{A}H^{(i)}W^{(i)})$

$H^{(i+1)} = \sigma\left(\hat{A}^0 H^{(i)} W_0^{(i)} \middle| \hat{A}^1 H^{(i)} W_1^{(i)} \middle| \dots\right)$

(a) Traditional graph convolution. (b) Our mixed feature model.

MixHop: Higher-Order Graph Convolutional Architectures via Sparsified Neighborhood Mixing

MixHop Model

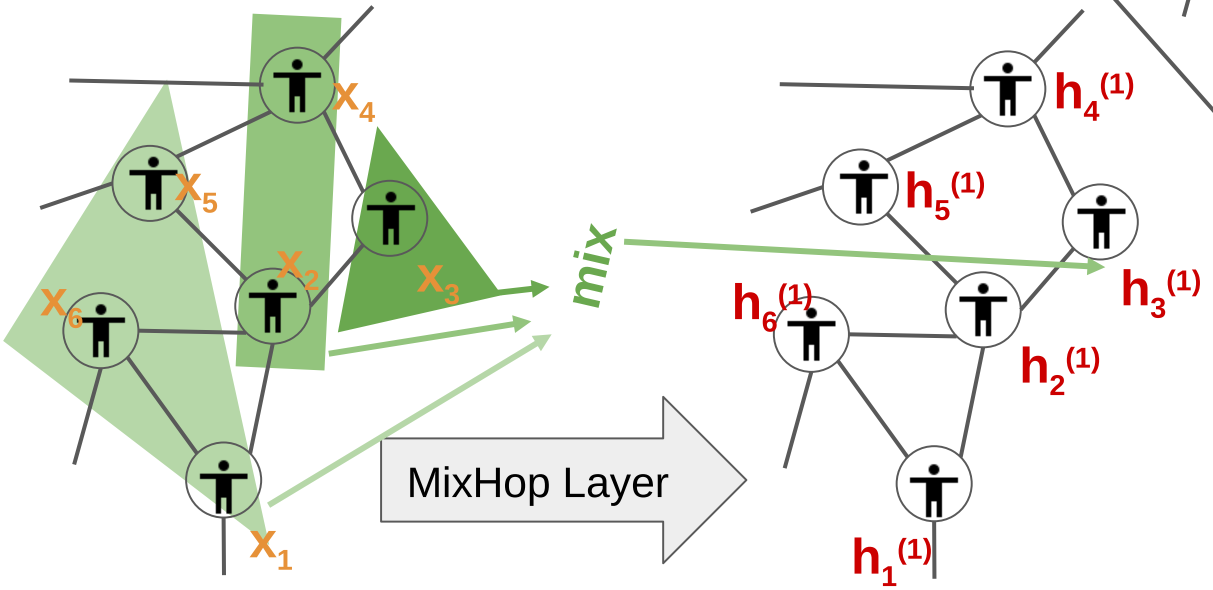


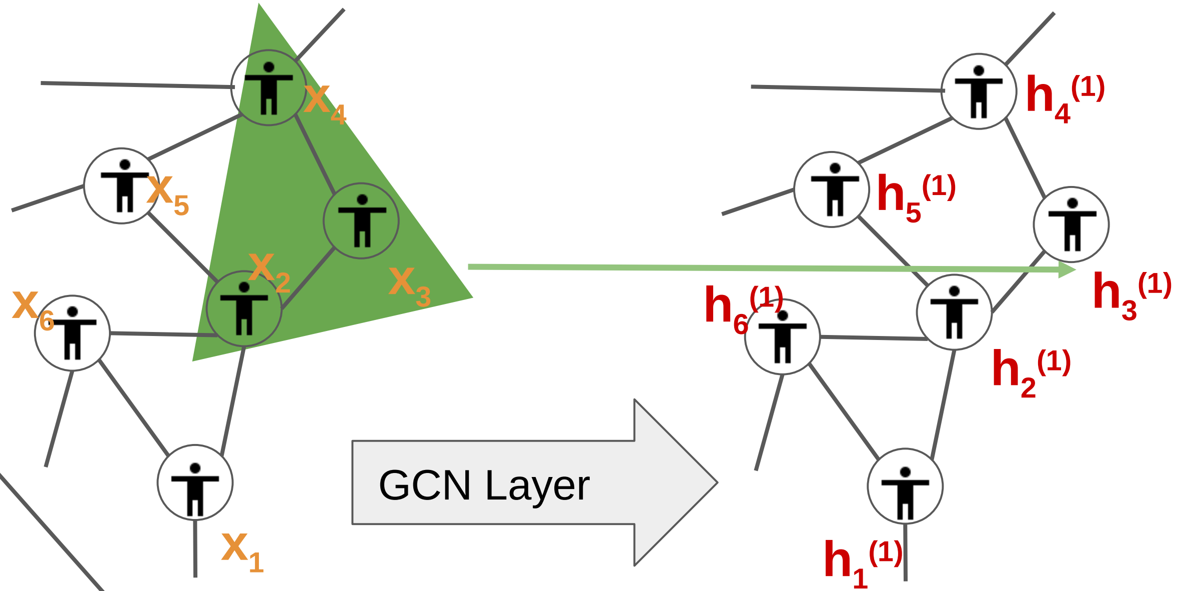
MixHop: Higher-Order Graph Convolutional Architectures via Sparsified Neighborhood Mixing

MixHop Model

😀 Can incorporate distant nodes

😀 Can mix neighbors across distances
i.e. **can learn Gabor Filters**!





😢 Appendix Experiments of [7]
shows no gains beyond 2 layers

😢 cannot arbitrary mix neighbors
from various distances
→ **cannot learn Gabor Filters**!

Vanilla GCN Model

Addressing Pitfall #2: Scalability

Two approaches:
1- Sampling-based
2- Decoupling-based (aka linearized models)

Addressing Scalability via Sampling

Rather than processing graph as a whole, at each training step, sample a few nodes and edges.

- [uniform] sample training nodes **uniformly at random**, its edges **uniformly at random**
- [biased] e.g., sample high degree nodes with higher probability.
- [learnable] sample from trainable distribution.

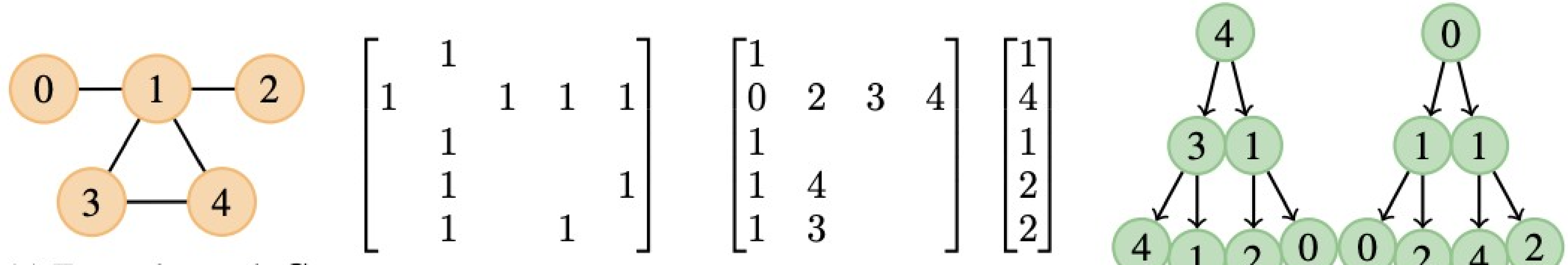
Papers: “GraphSAGE”, “GTTF”, “TF-GNN”, …

Papers “FastGCN”, “LADIES”, “ClusterGCN”, “GraphSAINT”, …

Papers: “DSKReG”, “Performance-adaptive sampling GNN”, “SubMix”

Graph Traversal with Tensor Functionals: A meta-Algorithm for Scalable Learning

- Graph Learning algorithms require the graph.
- They out to be **approximable** by graph traversals



(a) Example graph G

(b) Adjacency matrix for graph G

(c) CompactAdj for G with sparse $\widehat{A} \in \mathbb{Z}^{n \times n}$ and dense $\delta \in \mathbb{Z}^n$. We store IDs of adjacent nodes in $\widehat{A}$

(d) Walk Forest. GTTF invokes AccumulateFn once per (green) instance.

Figure 1: (c)&(d) Depict our data structure & traversal algorithm on a toy graph in (a)&(b).

WEDNESDAY: No class – I’ll do an in-person office hour instead
I’ll send an email reminder